High School Chemistry Syllabus

Course Title

High School Chemistry

Grade Level

Grades 9–12

Course Length

Full academic year

Credit

1 high-school science credit upon successful completion, subject to school graduation requirements.

Course Description

High School Chemistry is a teacher-directed, laboratory-based course examining matter and its interactions at the atomic, molecular, and macroscopic levels.

All students will progress through the curriculum together according to a common course schedule.

Students will use **Barron's Visual Learning: Chemistry** as the primary conceptual text. Teacher-created supplemental instruction will provide the mathematical skills, worked examples, practice problems, laboratory analysis, and additional content necessary for a complete high-school chemistry course.

Major topics include measurement, atomic structure, the periodic table, chemical bonding, molecular structure, chemical nomenclature, reactions, moles, stoichiometry, solutions, acids and bases, gases, thermochemistry, reaction rates, equilibrium, oxidation-reduction, and nuclear chemistry.

The course emphasizes direct instruction, mathematical problem solving, scientific reasoning, laboratory investigation, and cumulative understanding.

Primary Resource

Barron’s Visual Learning: Chemistry

Additional resources may include:

- River Tech High School Chemistry Supplement
- Teacher-created notes and problem sets
- ACS instructional resources
- PhET simulations
- OpenStax Chemistry reference material
- Laboratory investigations
- Demonstrations
- Additional scientific resources selected by the teacher

Course Format

This is a **traditional teacher-paced course**.

Students will:

- Study the same material at the same time.
- Follow a common unit schedule.
- Complete the same core assignments.
- Participate in scheduled class laboratories.
- Complete quizzes and tests on established dates.
- Meet common assignment deadlines.
- Move through the course as a class rather than independently.

Additional help or practice may be provided to students who need it, but the overall class pacing will remain common.

Students who are absent are responsible for making up missed instruction, assignments, labs, and assessments according to school policy.

Prerequisites

Students should have working knowledge of:

- Arithmetic
- Fractions
- Decimals
- Percentages
- Ratios and proportions
- Basic algebraic manipulation

Completion of or concurrent enrollment in Algebra I is recommended.

Course Objectives

Upon successful completion of the course, students should be able to:

- Use appropriate scientific measurements and units.
- Apply scientific notation.
- Apply significant figures.
- Perform dimensional analysis.
- Calculate density.
- Describe atomic structure.
- Interpret isotope notation.
- Calculate average atomic mass.
- Explain electron arrangement and periodic trends.
- Predict common ionic charges.
- Write and name ionic and molecular compounds.
- Construct basic Lewis structures.
- Apply VSEPR theory.
- Determine basic molecular polarity.
- Explain intermolecular forces.
- Interpret and balance chemical equations.
- Classify major reaction types.
- Calculate molar mass.
- Convert among mass, moles, and particles.
- Solve stoichiometric problems.
- Identify limiting reactants.
- Calculate theoretical and percent yield.
- Calculate solution concentration.
- Apply molarity and dilution relationships.
- Explain acid-base chemistry.
- Perform introductory pH calculations.
- Analyze a basic titration.
- Apply major gas-law relationships.
- Analyze heat transfer using specific heat and calorimetry.
- Explain endothermic and exothermic processes.
- Describe factors affecting reaction rates.
- Explain activation energy and catalysis.
- Describe chemical equilibrium.
- Apply introductory Le Châtelier reasoning.
- Identify basic oxidation-reduction processes.
- Explain basic nuclear decay and half-life.
- Collect and analyze laboratory data.
- Evaluate experimental error.
- Support conclusions with evidence and calculations.

Course Units

Quarter 1 — Foundations and Atomic Structure

Unit 1 — Measurement and Matter

- Scientific measurement
- SI units
- Scientific notation
- Significant figures
- Dimensional analysis
- Mass and volume
- Density
- Classification of matter
- Physical and chemical changes

Unit 2 — Atomic Structure

- Protons
- Neutrons
- Electrons
- Atomic number
- Mass number
- Isotopes
- Average atomic mass
- Basic nuclear structure

Unit 3 — Electrons and the Periodic Table

- Electron arrangement
- Valence electrons
- Periodic organization
- Atomic radius
- Ionization energy
- Electronegativity
- Periodic trends

Quarter 2 — Bonding and Chemical Reactions

Unit 4 — Chemical Bonding

- Ions
- Ionic compounds
- Covalent compounds
- Chemical formulas
- Chemical nomenclature
- Lewis structures
- Molecular geometry
- Molecular polarity
- Intermolecular forces

Unit 5 — Chemical Reactions

- Chemical equations
- Conservation of mass
- Balancing equations
- Reaction types
- Introductory oxidation and reduction

Quarter 3 — Quantitative Chemistry

Unit 6 — The Mole

- Avogadro’s number
- Molar mass
- Mole conversions
- Percent composition
- Empirical formulas
- Molecular formulas

Unit 7 — Stoichiometry

- Mole ratios
- Mass-to-mass calculations
- Limiting reactants
- Excess reactants
- Theoretical yield
- Actual yield
- Percent yield

Unit 8 — Solutions

- Solubility
- Solutes and solvents
- Electrolytes
- Concentration
- Molarity
- Dilution

Unit 9 — Acids and Bases

- Acid-base properties
- pH
- Neutralization
- Introductory acid-base calculations
- Titration

Quarter 4 — Gases, Energy, and Reaction Behavior

Unit 10 — Gases

- Pressure
- Temperature
- Volume
- Boyle’s law
- Charles’s law
- Combined gas law
- Ideal gas law

Unit 11 — Thermochemistry

- Heat and temperature
- Specific heat
- Calorimetry
- Endothermic reactions
- Exothermic reactions
- Enthalpy

Unit 12 — Reaction Behavior

- Collision theory
- Reaction rates
- Activation energy
- Catalysts
- Chemical equilibrium
- Le Châtelier’s principle

Unit 13 — Nuclear Chemistry and Course Review

- Radioactivity
- Nuclear decay
- Half-life
- Cumulative chemistry review
- Final assessment

Typical Class Structure

A typical class period may include:

1. Bellwork or review
2. Teacher-led instruction
3. Barron’s visual reference
4. Worked examples
5. Guided practice
6. Independent practice
7. Laboratory, simulation, or application activity when scheduled
8. Assignment or exit check

The teacher will determine when the class moves to the next topic.

Homework and Practice

Students should expect regular chemistry practice.

Assignments may include:

- Reading
- Guided notes
- Problem sets
- Calculations
- Vocabulary
- Written explanations
- Lab preparation
- Lab analysis
- Review assignments

Chemistry requires regular practice, particularly in quantitative units.

Assignments will have common due dates.

Laboratory Program

Students will complete selected laboratory investigations as a class throughout the year.

Possible investigations include:

- Density and measurement
- Separation of mixtures
- Solubility and polarity
- Conservation of mass
- Reaction rate
- Acid-base chemistry
- Calorimetry
- Stoichiometry and percent yield
- Titration
- Gas behavior

Laboratory activities may be replaced with demonstrations or simulations when necessary for safety, equipment limitations, or instructional effectiveness.

Laboratory Reports

Depending on the investigation, students may be required to include:

- Purpose
- Hypothesis or prediction

- Procedure
- Observations
- Data
- Calculations
- Graphs
- Sources of error
- Conclusion

Not every laboratory activity will require a complete formal report.

Assessment

Students may be evaluated through:

- Homework
- Classwork
- Problem sets
- Quizzes
- Unit tests
- Laboratory work
- Laboratory reports
- Projects
- Quarter assessments
- Cumulative final examination

Grading Policies

Quarterly grades are derived from two distinct components: **Academic Performance** (evaluated assignments) and **Participation** (teacher-assessed engagement and preparation). While most courses utilize an equal 50/50 weighting, specific courses may be designated as Skill-Heavy (60/40) or Participation-Heavy (40/60) at the teacher's discretion.

The Academic component is calculated by converting each assignment into a percentage and a corresponding letter grade. This figure represents a combination of the overall quarterly trend (60%) and the average of the five most recent assignments (40%). This structure ensures that student improvement in the latter half of the quarter significantly influences the final grade.

Every assessment carries equal weight within the grading system. Small quizzes and major projects exert the same influence on the final average, emphasizing the importance of consistent performance over any individual result.

Grades are recorded using a letter-based system where percentages are snapped to established midpoints. Consequently, numerical differences within the same letter-band (such as 84% and 86%) result in the same grade. Students should focus on achieving higher letter-bands rather than individual points.

Missing assignments will be recorded as a zero after the designated grace period. Until that time, the displayed grade only reflects completed work, which may not accurately represent a student's current academic standing.

Grading Scale: A+ 97–100 · A 93–96 · A– 90–92 · B+ 87–89 · B 83–86 · B– 80–82 · C+ 77–79 · C 73–76 · C– 70–72 · D+ 67–69 · D 63–66 · D– 60–62 · F below 60

The final year grade is weighted: Q1 20%, Q2 25%, Q3 25%, and Q4 30%. Cumulative GPA is calculated on a 4.3 scale, with an A+ representing a 4.3.

Tests and Quizzes

Quizzes will normally assess material recently taught.

Each major unit will generally conclude with a test or other formal assessment.

Major assessments may include material from earlier portions of the course because chemistry concepts build upon one another.

Students will normally receive structured review before major tests.

Mathematics Expectations

High-school chemistry requires students to show their mathematical reasoning.

Unless otherwise directed, calculations should include:

1. Appropriate formula or relationship
2. Given information
3. Substitution
4. Units
5. Mathematical work
6. Final answer

Answers without supporting work may receive reduced credit.

Scientific calculators may be used when permitted by the teacher.

Laboratory Safety

Students must:

- Follow teacher instructions at all times.
- Wear required protective equipment.
- Use chemicals only as directed.

- Read labels before using materials.
- Never taste laboratory substances.
- Never conduct unauthorized experiments.
- Maintain a clean work area.
- Properly dispose of materials.
- Immediately report spills, injuries, or broken glass.
- Follow all instructions involving heat and chemicals.

Unsafe behavior may result in immediate removal from the laboratory activity and additional consequences according to school policy.

Academic Integrity

Students are expected to complete their own assessed work.

Students may collaborate when specifically permitted during class activities or laboratories, but individual assignments, calculations, quizzes, and tests must represent each student's own understanding unless otherwise instructed.

Unauthorized copying, answer sharing, or use of outside resources during assessments will be handled according to school policy.

Student Materials

Students should bring:

- Barron’s Visual Learning: Chemistry
- Chemistry notebook or binder
- Pencil
- Scientific calculator
- School-issued device when requested

Student Expectations

Students are expected to:

- Attend class regularly.
- Arrive prepared and on time.
- Participate in instruction.
- Take organized notes.
- Complete assignments by their deadlines.
- Practice chemistry calculations regularly.
- Ask questions when help is needed.
- Follow laboratory safety requirements.
- Work professionally with classmates.

- Prepare for quizzes and tests.

Final Assessment

The course will conclude with a cumulative final assessment.

The final may include:

- Chemistry concepts
- Vocabulary
- Chemical equations
- Calculations
- Data analysis
- Laboratory interpretation
- Application of concepts from across the year

A practical or laboratory component may also be included.

Final Course Goal

By the end of High School Chemistry, students should be able to connect:

observable chemical behavior

↓

atomic and molecular explanations

↓

chemical symbols and equations

↓

quantitative calculations

Students should leave the course prepared for future high-school or introductory college science coursework.